

Triasha Sarkar

Machine Learning Engineer | Ex-Rolls-Royce · United States · tsarkar34@gatech.edu · [LinkedIn](#) · [GitHub](#) · [Portfolio](#)

SUMMARY

Machine learning engineer with aerospace systems experience building evaluation pipelines, retrieval systems, surrogate models, and deployment evidence. I work across data contracts, model evaluation, reproducible experiments, and production boundaries. My focus is making ML results measurable, reviewable, and useful to engineering teams.

EXPERIENCE

Systems Engineer — Alten India / Rolls-Royce engagement Oct 2023 – Apr 2025

- Worked on aircraft-engine systems development and certification activities for the Trent XWB-84 program.
- Collected, analyzed, and managed customer and system requirements using IBM DOORS.
- Supported system architecture, verification and validation, integration, risk tracking, and compliance documentation.
- Worked with engineering, customer, and program stakeholders to keep requirements and evidence aligned.

Graduate Researcher — Georgia Tech Aerospace Systems Design Laboratory 2025 – 2026

- Worked on aviation systems, scientific ML, sustainable aviation, simulation reliability, and decision-support tools.
- Built reproducible research software for retrieval, ranking, surrogate modeling, evaluation, and deployment studies.
- Translated model behavior and limitations into engineering-facing evidence and documentation.

Project EAGLE — Vehicle Grand Challenge, Georgia Tech 2025 – 2026

Developed a vehicle sizing and synthesis tool for an eco-friendly airborne vehicle, including requirements, cabin geometry, passenger layout, and system-level trade studies.

Project GREEN TEA — System of Systems Grand Challenge, Georgia Tech 2025 – 2026

Developed a techno-economic and impact-analysis framework for forest-residue bioenergy and sustainable aviation fuel, including jobs, emissions, and forest-health modeling.

SELECTED ENGINEERING PROJECTS

AeroRAG-X — Retrieval and agent evaluation [GitHub](#)

Built evaluation contracts for retrieval, citation support, agent trajectories, safe degradation, observability, and held-out comparisons. Includes QASPER/SciFact evaluation paths and citation-corruption tests.

NewsLens — Stateful news retrieval service [GitHub](#)

Built a containerized retrieval service with separate application and state hosts, Kafka, Postgres, health checks, evidence runners, and documented deployment boundaries.

IntegrityBench — Integrity and safety evaluation [GitHub](#)

Built fail-closed release gates, public evidence receipts, CUDA transformer evaluation paths, and candidate assessment tooling without overstating missing human-review evidence.

EdgeGenBench — Edge inference evidence [GitHub](#)

Built reference and QNN evidence validators, generated-data aircraft design experiments, runtime contracts, and device-evidence capture workflows.

SELECTED ENGINEERING PROJECTS (CONTINUED)

AeroSynth-Eval — Generative-model evaluation [GitHub](#)

Built preference, calibration, transfer, downstream-utility, grouped-uncertainty, and human-alignment evaluation contracts. Synthetic and real-data claims are kept separate.

Equity-Backtest — Leakage-aware research pipeline [GitHub](#)

Built a point-in-time backtesting pipeline with membership validation, leakage checks, turnover, capacity constraints, and explicit boundaries around licensed CRSP data.

Surrogate Model Learning — Robust regression [GitHub](#)

Built Gaussian-process and regression workflows for engineering data, repeated grouped robustness reports, uncertainty checks, and operating-domain warnings.

TECHNICAL SKILLS

Languages: Python, C++, Java, SQL, MATLAB, Bash

ML: PyTorch, scikit-learn, Gaussian processes, regression, retrieval, ranking, evaluation, uncertainty, time series

Systems: Docker, Kafka, Postgres, AWS, CI, ONNX, QNN, REST APIs, observability, reproducibility contracts

Engineering: requirements management, systems architecture, verification and validation, risk analysis, model cards, technical documentation

RESEARCH AND ENGINEERING FOCUS

- Scientific machine learning and surrogate modeling for aerospace and energy systems.
- Evaluation of retrieval, generation, safety, citation support, and model integrity.
- Reliable ML systems with explicit data, deployment, and evidence boundaries.

EARLIER AEROSPACE PROJECTS

- **GKN Aerospace hybrid aircraft design:** thermal-management integration for a zero-emission demonstrator using hybrid, electric, and hydrogen system concepts.
- **Supersonic business-jet thesis:** studied primary-nozzle geometry and noise-mitigation constraints for take-off and landing.
- **Bird-strike structural analysis:** used ANSYS to compare nose-cone material response under impact loading.
- **Rocket test-pad internship, STAR:** contributed to static-test-pad mechanical design and Arduino-based avionics and safety systems.
- **Aircraft structures internship, BAIL:** analyzed stress and material behavior under environmental loading and supported experimental test setups.

EDUCATION

M.S., Aerospace Engineering — Georgia Institute of Technology 2026

M.Sc., Aerospace Vehicle Design — Cranfield University 2023

B.Tech., Aerospace Engineering — SRM Institute of Science and Technology 2021